

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. CHB-601: Analytical Chemistry-I

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) What are the units of molar absorptivity (ϵ) and specific absorbance (a)?
 - (b) Which chromatographic technique is a two-dimensional chromatography? Give one example.
 - (c) Define the term retention time (t_R) in chromatography.
 - (d) Beer's law is obeyed in the case of an aqueous solution of a weak acid when the solution is made strongly acidic or basic. Explain why.
 - (e) The distribution constant (K_D) for iodine between an organic solvent and water is 85. Calculate the concentration of I_2 remaining in water after extraction.
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Question 2.

- (a) Show that for n successive extractions with volume V_x of organic solvent, the fraction of solute remaining in aqueous phase of volume V_{aq} is:

$$q = \left(\frac{V_{aq}}{V_{aq} + DV_x} \right)^n$$
[8]

- (b) Forty milliliters of 0.20 M butyric acid in water is extracted with 20 mL of ether. After the layers are separated, it was found that 1.0 mmol of butyric acid remained in the aqueous layer. Calculate the distribution ratio (D). [6]
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Question 3. Illustrate the following solvent extraction techniques: [3 × 4.5 = 13.5]

- (a) Multiple extractions vs. single extraction
 - (b) Synergic solvent extraction
 - (c) Ion-pair extraction of metal chelates
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Question 4.

- (a) A compound X exhibits a molar absorptivity of $2.45 \times 10^4 \text{ L mol}^{-1} \text{ cm}^{-1}$ at 450 nm. What concentration of X in a solution causes a 25% decrease in the radiant power at 450 nm in a 1.0 cm cell? [6]
- (b) What is a Ringbom plot (or Ayres plot)? Discuss its advantages and limitations in spectrophotometric analysis. [8]
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Question 5.

- (a) Discuss photometric errors in spectrophotometry and how they vary with transmittance (T). Prove that error is minimum when $T \approx 0.368$. [8]
- (b) Illustrate the principles of one-component and two-component analysis using UV-Visible spectrophotometry. [6]